How Nanogenerator Band Can Heal Wounds Faster

The triboelectric nanogenerator (TENG) is a fascinating technology that dramatically speeds up healing. It reduces healing time to one-fourth of the normal healing process. It can heal even nasty wounds of a diabetic patient. Moreover, the bio-degradable triboelectric nanogenerator (BD-TENG) is, as the name suggests, biodegradable. This means there is no need to open up the patient's body for taking out the implant

Rathindra Nath Biswas is a 1964-batch chemical engineering graduate from Jadavpur University, Calcutta. He was awarded a certificate for designing Benzol Plant by Giprokoks, USSR, and Certificate of Honour by Indian Institute of Metals. He has published 35 research papers in various journals. He retired from service as head - MECON, Durgapur

At present there are 73 million diabetes patients in India (world figure is 406 million), out of which fifteen per cent develop foot ulcers, which are difficult to heal. Moreover, accident victims also receive very nasty wounds. Most of these patients require hospitalisation for treatment, wherein electrotherapy is often applied using bulky machines and complicated wiring. However, these machines, if not controlled properly, could generate oxygen radicals inside the wound, which may cause cancer.

Recently, a low-cost wound dressing has been developed in the US that could dramatically speed up healing. The method converts the patient's own body motion into electrical energy with the help of a wearable nanogenerator band and applies gentle electrical pulses at the site of an injury. In laboratory tests, the bandage with nanogenerator reduced healing time to only three days as compared to nearly two weeks for the normal healing process.

Acute and chronic wounds represent a substantial burden on healthcare worldwide. Use of electrical stimulation in this method lowers the financial burden of the patient. The device is as convenient as putting a bandage on the skin.

This new dressing consists of small electrodes placed near the injury site. The electrodes are linked to a band holding energy-harvesting units called nanogenerators, which are looped around the wearer's torso. The natural expansion and contraction of the wearer's ribcage during breathing powers the nanogenerators, which deliver low-intensity electric pulses.

Nature of these pulses is similar to the way the human body generates an internal electric field. In fact, these low-power pulses do not harm healthy tissues like traditional high-power electrotherapy devices might. Also, exposing cells to high-energy electrical pulses causes them to produce almost five times more reactive oxygen species—a major risk factor for cancer and cellular aging compared to cells that are

Fig. 1: Wearable electric bandage

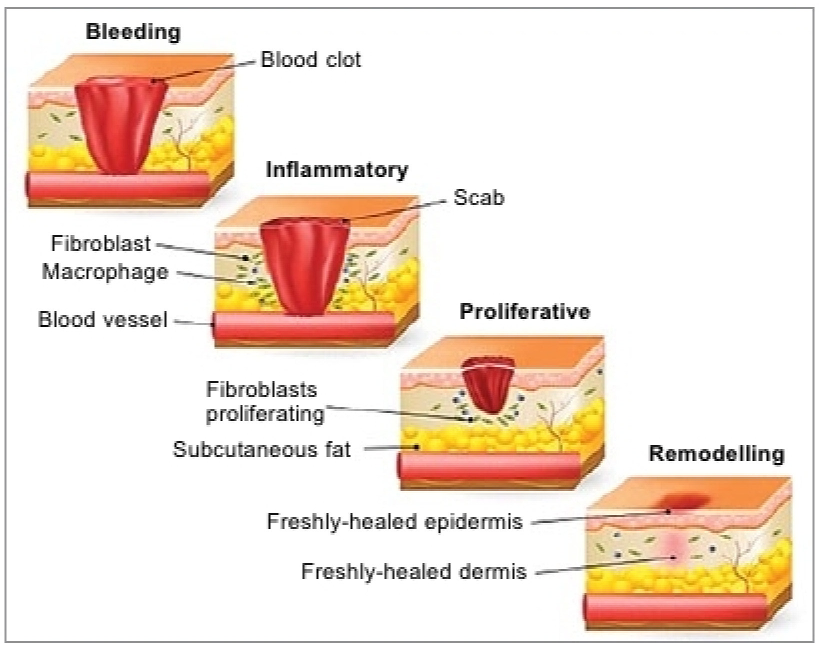

Fig. 2: Ionic basis of the wound's electrical response